Quantum Circuit Simulation

Riccardo Mattia

1 Introduction

Simulating quantum systems on classical hardware involves managing a state vector that grows exponentially with the number of qubits n. This project implements a simulator capable of executing standard gates and parametric rotations while providing benchmarking tools to analyze performance scaling.

2 Theoretical Framework

A quantum register of n qubits is represented as a complex probability amplitude vector in a 2^n-dimensional Hilbert space:

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i |i\rangle, \quad \text{where} \sum_{i=0}^{2^n-1} |\alpha_i|^2 = 1 \tag{1}$$

Quantum gates are modeled as unitary operators acting on this state vector. The following gates have been implemented in the simulator:

2.1 Single-Qubit Gates

Single-qubit gates are 2×2 unitary matrices that modify the state of a target qubit.

- **Hadamard** (H): Creates a balanced superposition of basis states.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{2}$$

- **Pauli-X** (X): Acts as a quantum NOT gate, flipping $|0\rangle \leftrightarrow |1\rangle$.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{3}$$

- **Pauli-Z** (Z): Applies a π phase shift to the $|1\rangle$ state.

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4}$$

- **Rotation Gates** (R_x, R_y, R_z): Represent rotations around the Bloch sphere axes. For example, $R_y(\theta)$ is defined as:

$$R_y(\theta) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \tag{5}$$

2.2 Multi-Qubit Gates

Multi-qubit gates are essential for creating entanglement and performing conditional logic.

- **Controlled-NOT** ($CNOT$): Flips the target qubit if and only if the control qubit is in the $|1\rangle$ state.

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \tag{6}$$

- **Toffoli (CCNOT)**: A three-qubit gate that flips the target qubit if both control qubits are in the $|1\rangle$ state. This gate is universal for classical reversible computation.

2.3 Quantum Measurement

Measurement is a non-unitary operation that collapses the quantum state into a classical basis state $|i\rangle$ with probability $P(i) = |\alpha_i|^2$. In the implementation, the state vector is normalized post-measurement to ensure the remaining amplitudes satisfy the normalization condition.

3 Performance & Benchmarking

The simulator's primary challenge is the "Exponential Wall." As shown in Table 1, the execution time doubles for every additional qubit, consistent with $O(2^n)$ complexity.

[Image of exponential growth function graph]

Qubits (n)	State Vector Size	RAM Usage (approx.)	Execution Time (ms)
10	2^{10}	16 KB	2.43
20	2^{20}	16 MB	1,209.34
25	2^{25}	512 MB	49,385.70
30	2^{30}	16 GB	2,140,400.00

Table 1: Benchmark results for Hadamard gate application across the entire register.

4 Implementation Details

The core logic resides in the `QubitRegister` class. OpenMP is utilized to parallelize state updates, significantly reducing computation time on multi-core processors.

```
void QubitRegister::applyCNOT(int control, int target) {
    long long maskControl = 1LL << control;
    long long maskTarget = 1LL << target;

    #pragma omp parallel for
    for(long long i = 0; i < stateVector.size(); ++i) {
        if((i & maskControl) && (i & maskTarget)) {
            std::swap(stateVector[i], stateVector[i ^ maskTarget]);
        }
```

```
    }
}
```

Listing 1: Parallelized CNOT gate implementation

5 Case Study: GHZ State and Entanglement Validation

To evaluate the simulator's performance in handling multi-qubit entanglement, we implemented the Greenberger-Horne-Zeilinger (GHZ) state for $n = 3$ qubits:

$$|\Psi_{GHZ}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}} \tag{7}$$

This state serves as a rigorous test for both the computational engine and the visualization suite. The results are analyzed through two complementary perspectives.

5.1 Probability Distribution Analysis

The computational basis histogram (Figure 1) confirms the perfect correlation between the three qubits. The simulator shows only two dominant peaks at states $|000\rangle$ and $|111\rangle$, each with a probability $P \approx 0.5$. All other $2^n - 2$ states remain at zero, validating the correct application of the Hadamard gate followed by the CNOT entanglement chain.

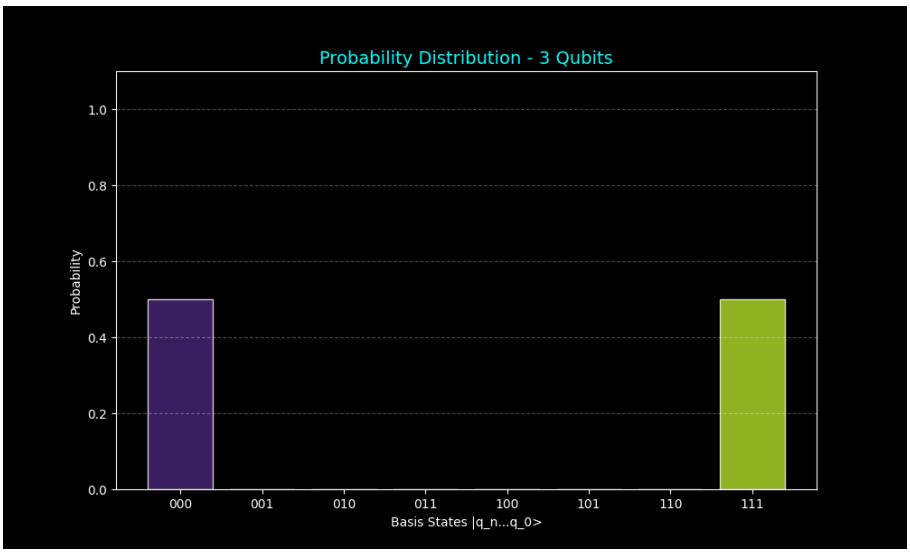

Figure 1: Probability distribution of the 3-qubit GHZ state showing maximum correlation.

5.2 Bloch Sphere Vanishing Vectors

A unique feature of entangled systems is revealed through the Bloch sphere projection. Unlike separable states, where vectors reach the sphere's surface, the GHZ state qubits exhibit "vanishing vectors" (Figure 2).

Since each qubit is maximally entangled with the rest of the register, its individual reduced density matrix $\rho_i = \mathrm{Tr}_{\hat{i}}(|\Psi\rangle\langle\Psi|)$ represents a maximally mixed state. Geometrically, this results in a state vector located at the origin of the Bloch sphere, signifying that no information can be gained by measuring a single qubit without considering the entire system.

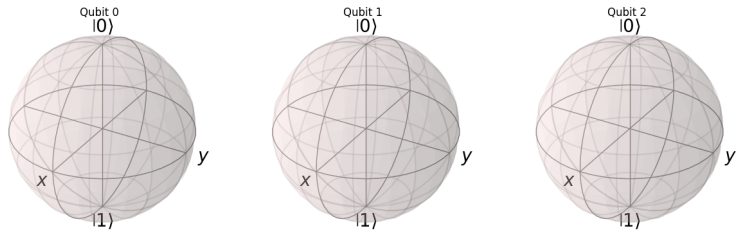

Figure 2: Bloch sphere projections for the GHZ state. The absence of visible vectors (state at the origin) provides visual evidence of maximal entanglement.

5.3 Multi-Qubit Bloch Projection

The developed visualization suite allows for the simultaneous analysis of multi-qubit registers. As shown in Figure 3, each qubit's state is isolated via partial trace, providing a clear geometric intuition of the register's configuration.

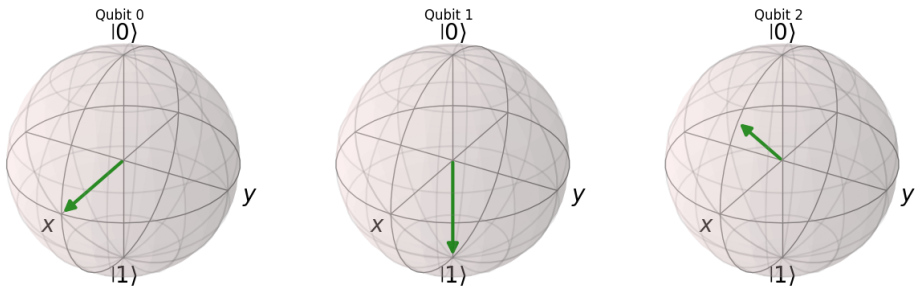

Figure 3: Bloch sphere projections for a 3-qubit system. Qubit 0 shows a $|+\rangle$ state, Qubit 1 is in the $|1\rangle$ state, and Qubit 2 represents a superposition with an arbitrary azimuthal phase.

6 Conclusion

The simulator successfully demonstrates the principles of quantum computation and the limitations of classical simulation.